

APP STORE REVIEW MINING FOR FEATURE IMPROVEMENT

A Data Mining Case Study

ABSTRACT

The rise of mobile applications has led to an abundance of user feedback through app store reviews. This study uses data mining and NLP techniques to analyze these reviews, uncovering patterns in user sentiment and app performance. Through ETL processing, text preprocessing, sentiment analysis with the VADER model, and app-wise clustering, the study transforms unstructured review data into actionable insights for data-driven feature improvement and enhanced user experience.

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1. INTRODUCTION

App stores are a primary medium for distributing software and gathering user feedback. Millions of reviews are written daily, providing direct insights into user experiences, satisfaction levels, and expectations. However, this data is largely unstructured and cannot be directly analyzed using traditional methods. Data mining techniques provide an efficient way to convert this unstructured text into structured information. By analyzing sentiment polarity and identifying common issues mentioned in reviews, developers can make informed decisions regarding app design, performance, and functionality improvements. In this study, user reviews from various mobile applications across categories such as food, social media, news, and games were collected from the Google Play Store. Using Python-based NLP and data mining tools, the reviews were preprocessed, analyzed for sentiment, and categorized into positive and negative clusters. The final outcome is a set of data-driven feature improvement recommendations for each app.

2. OBJECTIVES

- To extract and preprocess app store reviews for multiple applications using ETL methods.
- To perform sentiment analysis and classify reviews into positive, negative, and neutral categories.
- To cluster positive and negative reviews for each app separately.
- To identify the most frequently mentioned words in negative reviews.
- To automatically generate app-specific improvement suggestions based on sentiment patterns and complaint keywords.
- To visualize sentiment distribution across apps to compare performance and user satisfaction levels.

3. DATASET DESCRIPTION

The dataset used in this study was obtained from the Google Play Store User Reviews dataset available on Kaggle. Attributes included App, Translated_Review, Sentiment, Sentiment_Polarity, and Sentiment_Subjectivity. After importing the dataset, it contained over 64,000 reviews across multiple apps. Each record was preprocessed by converting text to lowercase, removing punctuation, special characters, and stop words, and lemmatizing words using WordNetLemmatizer. This cleaning ensured consistency and improved model accuracy.

4. METHODOLOGY

5.1 Data Extraction (E)

The dataset was extracted directly from the Kaggle dataset file `googleplaystore_user_reviews.csv`. Only relevant attributes such as App name, Translated Review, and Sentiment Polarity were selected for analysis.

5.2 Data Transformation (T)

Data transformation involved cleaning the review text, tokenizing and lemmatizing words, and removing stop words. The resulting cleaned text was stored in a new column called `Cleaned_Review`.

5.3 Data Loading (L)

Cleaned data was stored in a Pandas DataFrame for analysis. VADER sentiment analyzer was used to assign numeric sentiment scores and categorical sentiment labels.

5.4 Sentiment-Based Clustering

Reviews were grouped by App and clustered into positive and negative categories to identify aspects users liked and disliked.

5.5 Aspect Extraction

Top 10 most frequent words in each cluster were identified to reveal the main discussion topics using Python's Counter function.

5.6 Improvement Suggestion Generation

Based on frequent negative keywords, rule-based NLP logic was used to automatically generate app-specific improvement suggestions.

5.7 Visualization

Sentiment distribution per app was visualized using Seaborn bar plots to compare user satisfaction levels across different applications.

6. RESULTS AND ANALYSIS

The analysis produced a dataset linking each app to its top positive and negative keywords and automatically generated improvement suggestions. Below are sample results for a few applications:

App	Top Positive Words	Top Negative Words	Improvement Suggestion
Anthem Anywhere	information, claim, login, easy, app	log, update, slow, error, card	Multiple users report app crashes or freezing — prioritize debugging and stability improvements.

Apartments.com Rental Search	apartment, easy, finding, fix, useful	listing, review, search, rent, app	General user dissatisfaction noted — deeper analysis required for usability improvements.
Apex Launcher	update, launcher, icon, setting, version	setting, screen, drawer, cant, home	Recent updates seem to cause issues — improve QA before release.
Apple Daily	data, consumption, support, news	ad, crash, time, annoying	App stability and bug fixes needed to reduce crashes and unwanted ads.
Asphalt 8: Airborne	car, race, graphic, like	server, update, multiplayer, token	Fix server issues and optimize multiplayer stability.

Performance issues, advertisement complaints, and login difficulties were among the most common problems. Despite this, most applications retained positive overall sentiment. Visualization using Seaborn provided a clear comparison of sentiment trends across apps.

7. CONCLUSION

This study demonstrates how data mining and sentiment analysis can effectively extract actionable insights from user feedback. By combining ETL, OLAP, and rule-based NLP, developers can prioritize fixes, enhance performance, and improve user satisfaction. This approach is scalable and applicable to various domains involving textual feedback such as customer reviews and product feedback.

8. REFERENCES

1. Kaushal Trivedi, "Google Play Store User Reviews Dataset," Kaggle, 2018.
2. Hutto, C. & Gilbert, E. (2014). "VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text." ICWSM-14.
3. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, 2011.
4. Bird, Klein, & Loper, "Natural Language Processing with Python (NLTK)," O'Reilly Media, 2009.
5. Python Libraries: Pandas, Matplotlib, Seaborn, WordCloud, TextBlob, and VaderSentiment Documentation.